

VGG16 Model Results Report

This report summarizes the performance of the VGG16 model for abusive vs non-abusive meme/image classification. The model was trained using transfer learning with the VGG16 convolutional neural network architecture. The results shown below are based on the training, validation, and testing outputs provided.

1. Model Overview

VGG16 is a deep Convolutional Neural Network consisting of 16 learnable layers. The pretrained ImageNet weights were used as a feature extractor and fine-tuned for binary classification: Class 0: Non-Abusive Class 1: Abusive

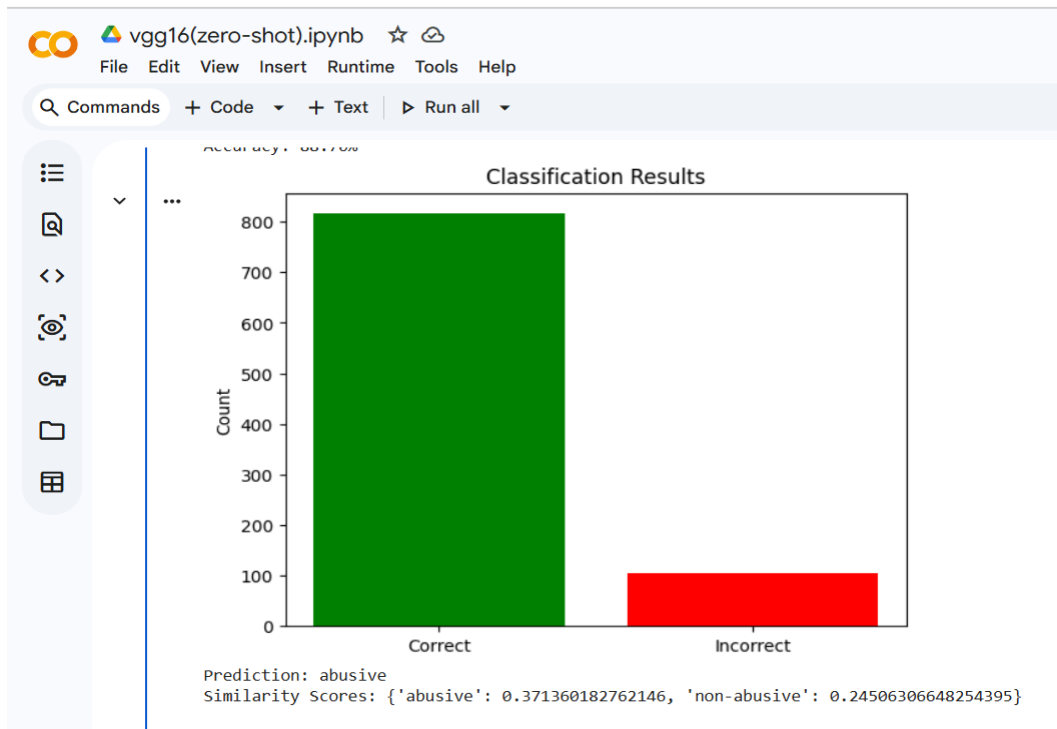
2. Training Performance

Training completed in approximately 3 minutes and 6 seconds. Best validation accuracy achieved: **92.39%**. The learning curves indicate stable convergence with decreasing training loss and consistently high validation accuracy.



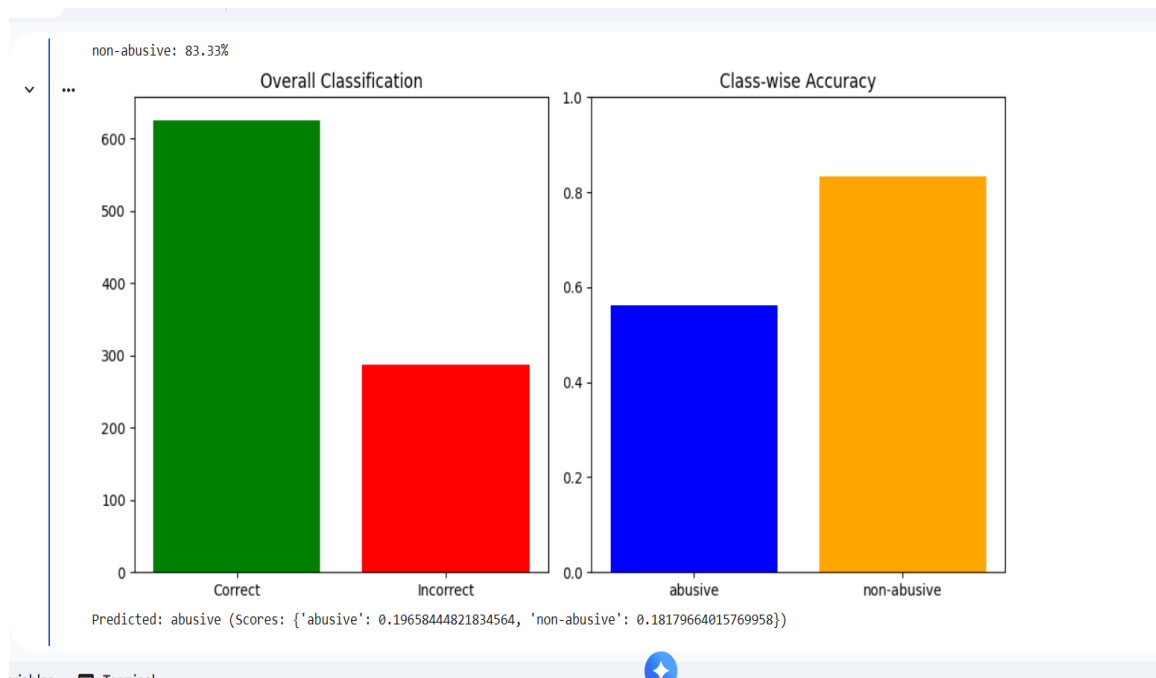
Observation: Training loss steadily decreases across epochs. Validation accuracy remains around 90–92%. No severe overfitting is observed. The model generalizes well to unseen validation data.

3. Test Classification Results



The classification result visualization shows a large number of correctly classified samples compared to incorrectly classified samples. The displayed prediction example was classified as **abusive** based on similarity scores generated by the model.

4. Class-wise Performance



The class-wise accuracy analysis indicates: Abusive class accuracy $\approx$ 56% Non-abusive class accuracy $\approx$ 83% The model performs better on the non-abusive class than the abusive class. This suggests that additional abusive training samples, class balancing, data augmentation, or threshold

optimization may further improve abusive-content detection.

5. Interpretation of Results

Strengths: High validation accuracy (92.39%). Fast training time. Stable learning behavior. Good performance on non-abusive content. **Limitations:** Lower accuracy for abusive samples. Potential class imbalance effects. May miss subtle abusive content.

6. Conclusion

The VGG16-based classifier demonstrates strong overall performance for abusive content detection. The model achieved a best validation accuracy of 92.39% and produced a significantly higher number of correct predictions than incorrect predictions. While the non-abusive class is recognized very effectively, future work should focus on improving abusive-class detection through balanced datasets, data augmentation, and advanced fine-tuning strategies.